

Diversify the Decisions, Not Just the Assets

A Practical Architecture for Dynamic Asset Allocation When Strategy Choice Is Uncertain

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Abstract

Diversification usually starts with assets. This paper asks whether it should also reach the decisions that choose them. Dynamic asset allocation can respond to changing market evidence without requiring a precise forecast of the next regime, but many practical rules are built on the same momentum intuition. A portfolio can therefore hold several strategies and still depend on much the same decision logic.

We examine diversification at the decision level: What does each rule own? When does it change? How much risk does it take? We use Autonomous Dynamic Asset Allocation (ADAA) as a worked example of a rule-governed portfolio. ADAA is modular rather than a timeless recipe. The five sleeves are one investable example, and their number is not a design target. We first compare the public source rules, then ask whether the practitioner adaptations preserve meaningfully different decision roles.

In a June 2008-July 2026 historical simulation on a common ETF sample, the five investable sleeves make meaningfully different portfolio decisions even when some return correlations are high. We also compare 16 public rules associated with the 2023 design using portfolio decisions alone, without return or performance statistics. The five-family set documented in the 2023 design ranks above most feasible five-rule combinations in that comparison, but the exercise also reveals redundancy inside the original design. The later practitioner weights sit within a broad near-optimal region rather than at a unique backtest optimum. ADAA does not beat the best individual sleeve in hindsight and does not protect in every stress episode. The evidence supports a narrower proposition: diversify reasonable decision rules, avoid precision unsupported by the data, make failure modes visible, and treat sleeves as replaceable components.

1. Start With a Response, Not a Forecast

Dynamic allocation starts from a simple ambition: take less risk when the evidence weakens without pretending to know exactly when the next regime begins.

Momentum is one practical answer. It replaces the question, 'What will happen next?' with a more observable one: 'Is this exposure still being rewarded, or has the evidence changed?'

Takeaway. *The aim is not to call the turn. It is to decide in advance how the portfolio will respond when the evidence changes.*

A systematic rule cannot see the future either. Its advantage is narrower and more useful: the response is specified before the next market story arrives. That makes the process easier to review, limits after-the-fact improvisation, and makes failures easier to identify.

In ADAA, 'autonomous' means rule-governed rather than discretionary: the response rules are specified in advance. It does not imply a self-learning or self-optimizing system, and it does not remove the need for human governance.

There are several plausible reasons trends can persist. Information reaches investors at different speeds. Institutions adjust large portfolios gradually. Economic and policy regimes can last for months or years

rather than days. Investors may underreact at first and extrapolate later. The argument in this paper does not depend on any one explanation.

Momentum is well documented in equities and across asset classes (Jegadeesh and Titman 1993; Moskowitz, Ooi, and Pedersen 2012; Asness, Moskowitz, and Pedersen 2013). Behavioral and information-diffusion models offer possible explanations for its persistence (Barberis, Shleifer, and Vishny 1998; Hong and Stein 1999).

That intuition has produced a large family of practical allocation rules. Faber uses simple trend rules to alter risky-asset exposure, while Antonacci combines relative and absolute momentum. Keller and collaborators add canary assets, defensive universes, breadth, volatility, and correlation inputs. Adaptive Asset Allocation takes a broader portfolio-construction view and changes portfolio inputs as market conditions change (Faber 2007; Antonacci 2017; Butler et al. 2012; Keller and van Putten 2012; Keller 2022; Keller and Keuning 2023).

Despite the different mechanics, the common discipline is to let observed market behavior change risk exposure rather than hold the same allocation through every environment.

Every adaptive rule also has to choose a speed. Too slow and it can give back protection. Too fast and it can get whipsawed. Defensive at the wrong moment, it can miss the first stage of a sharp rebound.

Momentum's success creates a second uncertainty: model choice. Once a useful idea is widely understood, it can be expressed through many reasonable lookbacks, universes, canary rules, defensive assets, breadth filters, and weighting schemes. The menu grows, but certainty does not. Part of the problem moves from forecasting the market to choosing among models: which reasonable expression of momentum will work best in the next regime?

That model-selection concern also appears in the author's private June 2023 university project report documenting an earlier ADAA design. The report is used only to document the problem formulation and historical strategy set as they stood at the time. Section 4 separates that history, including the later adoption of HAA, from the present empirical validation.

Momentum can therefore be useful and still create concentration: several rules built on the same successful idea may end up making much the same decision. More strategy names do not necessarily create more decision diversity. The practical question is whether the rules actually disagree when it matters.

2. Five Strategies Can Still Be One Decision

Imagine a portfolio with five dynamic-allocation strategies. On paper it looks diversified: five models, five rule sets, five names. Five is incidental in this example; the same question applies to any multi-rule portfolio.

Now imagine that all five depend heavily on momentum, turn defensive in roughly the same month, and return to risk after roughly the same signal. The investor owns five strategies but may still be making one underlying decision.

The analogy is familiar. Owning five technology stocks is not the same as owning five independent sources of risk. Strategy diversification can fail in the same way.

Return correlation is a useful first check, but it describes outcomes more directly than decisions. Two rules can have correlated returns while changing portfolios at very different times, or hold different assets while de-risking on the same date.

This framing builds on substantial precedent. Forecast-combination and thick-modeling research studies diversity across forecasts or models (Batchelor and Dua 1995; Granger and Jeon 2004; Aiolfi and Favero 2005; Pesaran, Schleicher, and Zaffaroni 2009). Portfolio research warns against overconfidence in precisely

estimated optimal weights (Michaud 1989; DeMiguel, Garlappi, and Uppal 2009), while practitioner work compares strategy correlations and overlapping positions (Ludlow 2022). The paper's narrower contribution is a transparent way to examine decision-level diversification in dynamic asset-allocation rules.

We describe each rule in three simple dimensions:

- **What?** Which assets does the rule choose?
- **When?** When does it change its portfolio?
- **How much?** How much risk or defensive exposure does it take?

These three questions turn the abstract idea of strategy diversification into observable behavior. In the Decision-Space comparison, target-weight distance captures What and How Much together, holdings overlap adds a separate composition check, and transition-date overlap captures When. Strategy names can change; the questions still work.

A slow rule can diversify a fast one even when their returns are moderately correlated. Two rules can own different assets yet de-risk on the same dates. A portfolio can therefore be diversified in assets and concentrated in its decision clock.

ADAA is diversified at the decision level only if its sleeves make meaningfully different decisions and the portfolio can retain that diversity as its components evolve.

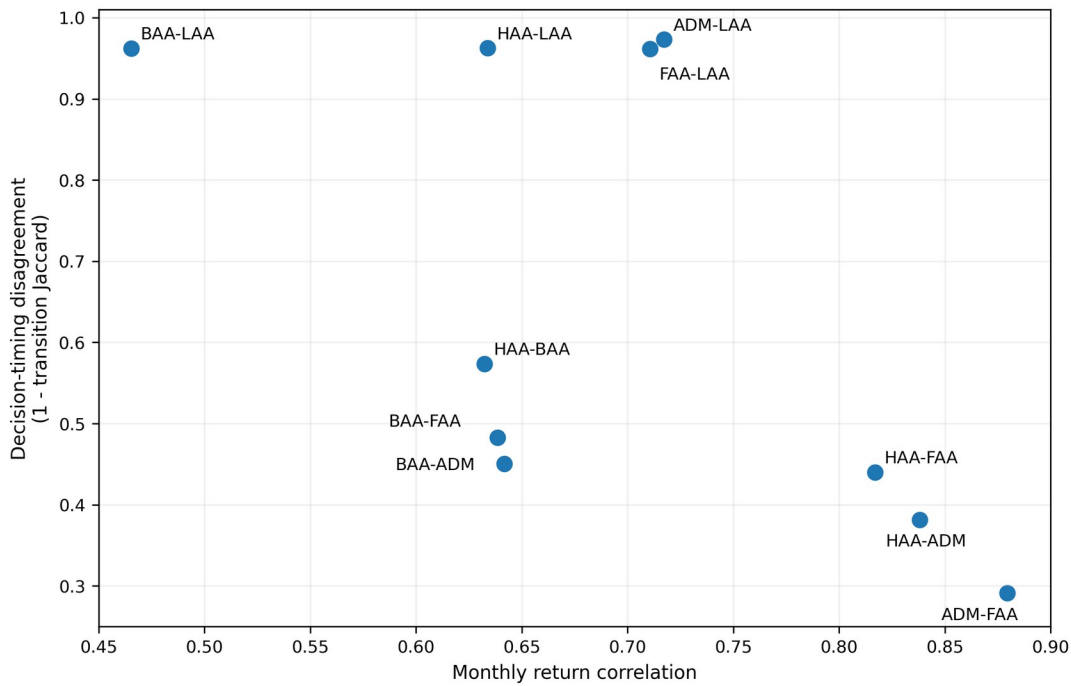


Figure 1. Similar returns do not imply similar decisions. The horizontal axis shows monthly return correlation. The vertical axis shows disagreement in portfolio-transition timing, measured as one minus the Jaccard overlap of their transition months. A value near one means the pair rarely changes its target weights in the same month. Return correlation remains useful, but it does not fully describe how rules behave.

3. A Strategy Is More Than Its Tickers

A published rule is a blueprint, not an investable portfolio. New ETFs appear; some original instruments are unavailable, expensive, or narrower than the intended exposure. The practical difficulty is that changing the assets used to implement the rule can also change the strategy itself.

That distinction matters because the current ADAA sleeves are not presented as untouched copies of published strategies. They are practitioner adaptations inspired by public rule families.

The comparison therefore asks three questions: What did the source rule actually do? What changed when the rule was adapted for ADAA? Did the adaptation preserve the role that made the source rule useful in the first place?

The trap is familiar: modify a rule or universe, observe a better history, and then use that same history as the justification for the modification.

The standard here is stricter: an adaptation is role-preserving only if it preserves the parent rule's underlying decision role; otherwise it is a redesign.

Table 1 summarizes the current classification.

Table 1. From published rules to investable sleeves

Sleeve	Public strategy family	Main ADAA adaptation	Classification	What the comparison shows
HAA	Keller & Keuning HAA	Wider offensive universe, Top-6, modified replacement logic	Substantial practitioner adaptation	The TIP-canary regime clock is preserved, but the breadth and composition of holdings change materially.
BAA Aggressive	Keller BAA Aggressive	Mainly VEA/VWO/BNL → EFA/EEM/AGG ETF substitutions	Close-to-parent adaptation	The current sleeve exactly matches the parent rule when applied to the ADAA ETF set; the source and ADAA ETF sets produce different states in 12 of 216 months.
ADM	Hanly Accelerating Dual Momentum	Narrow one-position selector → broad multi-asset Top-6 sleeve	Substantial redesign	The role itself is broadened; this is better viewed as a redesign than as a close adaptation of the source ADM rule.
FAA	Keller & van Putten FAA	Larger universe, 3/6/12 momentum structure, peer-only correlation, exact Top-6	Substantial rule and universe adaptation	Both rule and universe changes materially alter decisions.
LAA	Keller LAA	Persistent core retained; IWD allocation → SPY/EEM/EWY mix; timing information set clarified	Role-preserving adaptation	Holdings change, while the slow transition clock is largely preserved.

Source note: full source history, reasons for the modifications, and measured effects of moving from public parent rules to investable versions are provided in the public replication materials; the main-text table is intentionally compact.

A strategy is not its ticker list, but the ticker list is not irrelevant either. We therefore separate the choice of a public strategy family from the adaptation that turns it into an investable ADAA sleeve. Where possible, we first hold the parent rule fixed and change only the universe; then we hold the universe fixed and change the rule. Where that two-step comparison is not meaningful, as with the narrow parent ADM and the broad Top-6 multi-asset version, we do not force a mechanical 2×2 comparison.

Table 1B. Changes to the universe and rule can change the decision

Sleeve	Most direct comparison	Observed decision difference	Interpretation
HAA	Same parent rule: source universe vs ADAA universe	L1 = 0.563; timing agreement = 0.692	The wider universe changes the parent rule materially before Top-N/replacement changes are applied.
BAA Aggressive	Same parent rule: source vs ADAA ETF set	L1 = 0.275; timing agreement = 0.814	Mostly role-preserving, but ETF substitution changes some decisions.
ADM	Source rule design vs ADAA rule design	L1 = 2.000; timing agreement = 0.314	The redesign is too broad for a simple 2×2 comparison that changes one element at a time.
FAA	Same parent rule: source universe vs ADAA universe	L1 = 1.336; timing agreement = 0.686	Universe expansion alone changes decisions materially; later rule changes add another material change.
LAA	Same parent rule: source vs ADAA equity allocation	L1 = 0.500; timing agreement = 1.000	Holdings change while the parent transition clock is preserved exactly.

Measurement note: L1 is the average sum of absolute target-weight differences after mapping equivalent ETF exposures to common economic exposures (range 0-2). Appendix A divides this quantity by two for its 0-1 Decision-Space scale. Timing agreement is the Jaccard overlap of transition dates. No return statistic enters the table.

The LAA comparison is especially clean. Replacing the original equity allocation changes what the sleeve owns, but under the parent timing rule the QQQ/SHY switching state is unchanged across the common sample. The holdings change; the decision clock does not.

ADM is the opposite case. The current ADAA-ADM differs too much from the narrow parent strategy in its portfolio rules to be described as a minor adaptation. It is more accurately described as a practitioner redesign inspired by the source idea.

The purpose of the comparison is to identify each modification clearly and show what it changed, rather than describe every adaptation as closely matching its source.

4. Different Rules Need Different Jobs

In this paper, ADAA combines five sleeves: HAA, BAA Aggressive, ADM, FAA, and LAA. Five has no special status and is not part of the framework definition. It is simply the current portfolio. In plain English, the five strategies have different jobs:

- **HAA** — Uses a canary signal to turn defensive when the regime signal weakens and selects more broadly across offensive assets when risk is on.

- **BAA Aggressive** — An explicit offensive/defensive regime switch between risk-on candidates and a defensive universe.
- **ADM** — A fast relative-momentum Top-6 selector across a broad multi-asset universe.
- **FAA** — A broad cross-sectional sleeve combining momentum with volatility and correlation information.
- **LAA** — A deliberately persistent sleeve with a much slower switching clock.

These five strategies were not selected by an algorithm. They reflect practitioner judgment applied across publicly documented strategy families, with components chosen for distinct decision roles rather than to satisfy a target strategy count. The private June 2023 report documents three parts of that intent: reliance on rules rather than subjective market views; concern about common momentum dependence; and use of a slow-moving sleeve with explicit drawdown awareness. Because the report was private, it documents the design as it stood at the time; it is not preregistration or forward validation. Some ticker substitutions and later replacements are less completely documented and rely partly on the author's recollection.

The original judgment was that spreading reliance across several reasonable rules was more defensible than concentrating in whichever backtest looked best. That explains the design; it does not validate it. The five sleeves are one investable portfolio, not a required five-part recipe. Components may be added, removed, revised, or replaced when evidence on role, redundancy, robustness, and practical implementation justifies a change.

***Takeaway.** The sleeves can change, and their number can change as well. The architecture is the part meant to survive.*

A fair criticism is that ADAA may be only a hand-built collection of familiar rules. We test that concern in three stages:

1. Reconstruct public source rules associated with the 2023 design and compare them without using performance data.
2. Compare how each source rule was adapted into an investable sleeve.
3. Evaluate the five investable ADAA sleeves together as a portfolio.

This ordering avoids a circular argument in which differences created by later modifications are then used as evidence that the original family selection was diverse.

The comparison reveals an important trade-off. Using the same return-free Decision-Space measure, the five public source rules represented in the 2023 design have an average pairwise distance of about 0.82. Their investable ADAA versions average about 0.64, roughly 22% lower.

The same pattern appears for the five rules used in the current portfolio: about 0.85 for the public parent rules versus 0.66 for the investable versions. Much of the compression comes from ADM and FAA, whose broad multi-asset redesigns account for much of the decline in pairwise distance.

That compression is not automatically a design failure. A sleeve can become more diversified internally, less concentrated, or easier to implement while becoming more similar to the other sleeves. The portfolio problem therefore has two levels: diversify within each strategy, while preserving useful disagreement between strategies.

The measure is meant to make hidden concentration visible as components change; it is not an automated strategy-selection rule.

Figure 2 summarizes the five current sleeves as decision fingerprints. The horizontal axis captures what by showing the average number of active holdings. The vertical axis captures when by showing how often target weights change. Marker size represents the average rule-defined defensive allocation, a simple view of how much of the portfolio the rule shifts toward defensive assets.

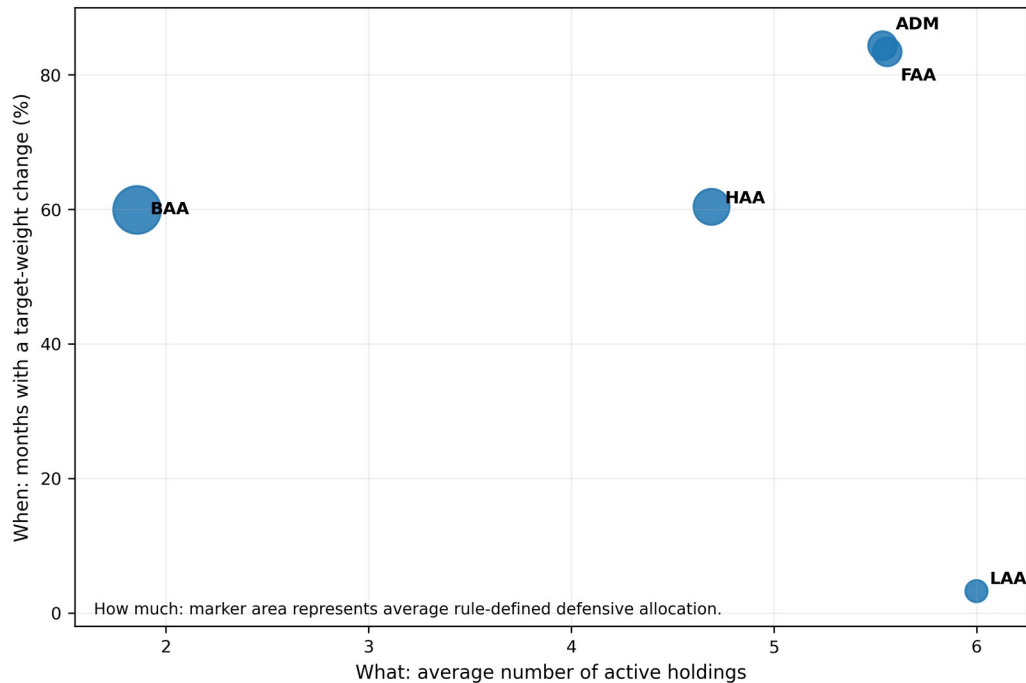


Figure 2. Decision fingerprints: What, When, and How Much. The dimensions are descriptive rather than a single optimized score. Their purpose is to reveal which kinds of disagreement each sleeve adds.

Figure 2 makes LAA's distinct clock easy to see. It does not make LAA a crash hedge: in the observed sample it did not always reduce drawdowns and was a material drag in 2022. Its contribution is better described as persistence - a different decision clock - than as guaranteed protection.

5. How the Current ADAA Portfolio Works

The process is deliberately compact so that its mechanics can be inspected and challenged. These rules describe the ADAA portfolio studied here, not a permanent specification.

1. Each sleeve observes its own signals at month-end.
2. Each sleeve produces a transparent target portfolio.
3. The five target portfolios are combined at the underlying-asset level.
4. New month-end targets become effective in the next holding period.
5. Trading costs are applied after offsetting sleeve-level trades at the final account level. A buy in one sleeve and a sell in another are therefore not charged separately when they cancel each other.

To compare the sleeves without letting the chosen sleeve weights drive the result, we use equal 20% weights. We also report the later practitioner weights used by the author: HAA 25%, BAA Aggressive 15%, ADM 17.5%, FAA 17.5%, and LAA 25%. These are not the June 2023 weights, and no written record from that time documents the exact rule used to choose them. Section 7 therefore treats them as a practical comparison, not as weights specified in advance for the study or as an optimal allocation.

A robust architecture should not require too many uncertain choices to be exactly right at once. If success depends on identifying both the future winning rule and its exact weight, the design is asking too much of foresight.

The next sections therefore separate two uncertainties: which reasonable rule will work best, and how precisely the rules need to be weighted.

6. Do the Rules Actually Disagree?

They do, but unevenly. That is why disagreement has to be measured rather than assumed.

Some pairs are clearly close. ADM and the current FAA sleeve have a gross monthly return correlation of about 0.88, the highest among the five sleeves, and their average target-weight distance is also relatively small. They are the clearest pair to examine for redundancy. We do not use that finding to replace either sleeve after seeing the results; any future replacement test should be defined before its results are known and evaluated separately.

Other pairs show why return correlation is not enough. HAA and LAA have a monthly return correlation of about 0.63, but their transition-timing disagreement is about 0.96. ADM and LAA have a return correlation above 0.71, yet their timing disagreement is about 0.97. The rules can have correlated returns while changing at very different times.

Returns describe realized outcomes, while transition overlap reveals part of the decision process behind them. The two measures answer different questions, and Figure 3 makes the timing dimension concrete.

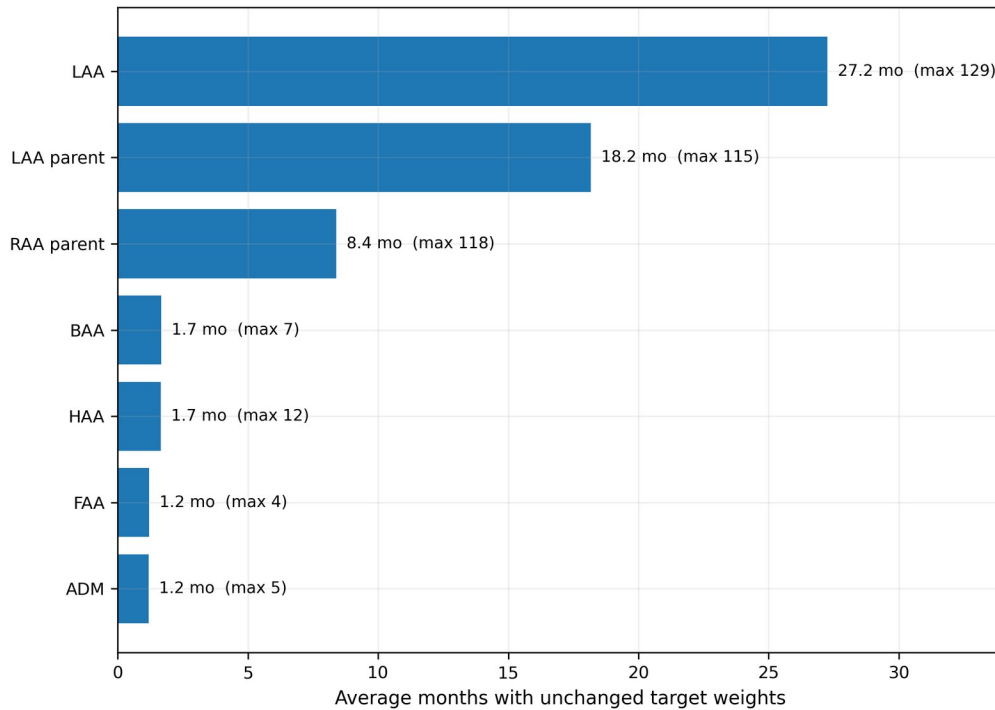


Figure 3. Different rules operate on different decision clocks. The current HAA and BAA targets remain unchanged for roughly 1.7 months on average; ADM and FAA for about 1.2 months. Current ADAA-LAA remains unchanged for about 27 months on average. The source LAA and the out-of-portfolio RAA comparator are also materially more persistent than the faster dynamic sleeves.

RAA is useful as an additional check. It is not an independent strategy family because it also comes from the Keller family of rules. But it shows that slow decision making is not unique to the ADAA-LAA sleeve used here. Similar persistence in source LAA and RAA supports the broader idea that a slow-moving sleeve can add a distinct decision-speed dimension.

That does not mean every multi-rule portfolio needs LAA, or that persistence is always beneficial. Persistence is a portfolio property, not a permanent job assignment. A future portfolio can seek the same property through a different sleeve if the evidence warrants it.

Even after the decision roles are visible, another uncertainty remains: which reasonable rule will perform best next? Figure 4 shows why hindsight is a difficult standard. The identity of the best standalone sleeve changes as the trailing window moves. HAA is the most frequent 60-month Sharpe winner, BAA also leads for long stretches, and LAA briefly leads in earlier windows. ADM and FAA never lead on this particular 60-month metric.



maximum gross Sharpe, while equal 20% weights achieve about 95.2%. Across 100,000 weight vectors that satisfy the minimum-weight constraint, about 40% achieve at least 95% of the ex-post maximum Sharpe.

Many materially different feasible weight vectors retain performance close to the ex-post maximum.

The estimated optimum also moves. In rolling 60-month optimizations, HAA's optimal weight ranges from the 10% floor to the 60% cap; BAA does the same; LAA ranges from the floor to above 50%. Block-bootstrap resampling also produces wide ranges for HAA, BAA, and LAA.

Figure 5 combines the two ideas. Panel A shows the observed sleeve-weight ranges among the 40,329 of 100,000 feasible Monte Carlo portfolios that retain at least 95% of the full-sample maximum Sharpe. It also shows the later practitioner allocation and the full-sample ex-post maximum-Sharpe weights. Panel B shows how the ex-post optimum moves under block resampling and rolling 60-month estimation. The dashed line marks the 10% minimum sleeve-weight constraint. The later practitioner weights are neither the full-sample optimum nor a special narrow optimum.

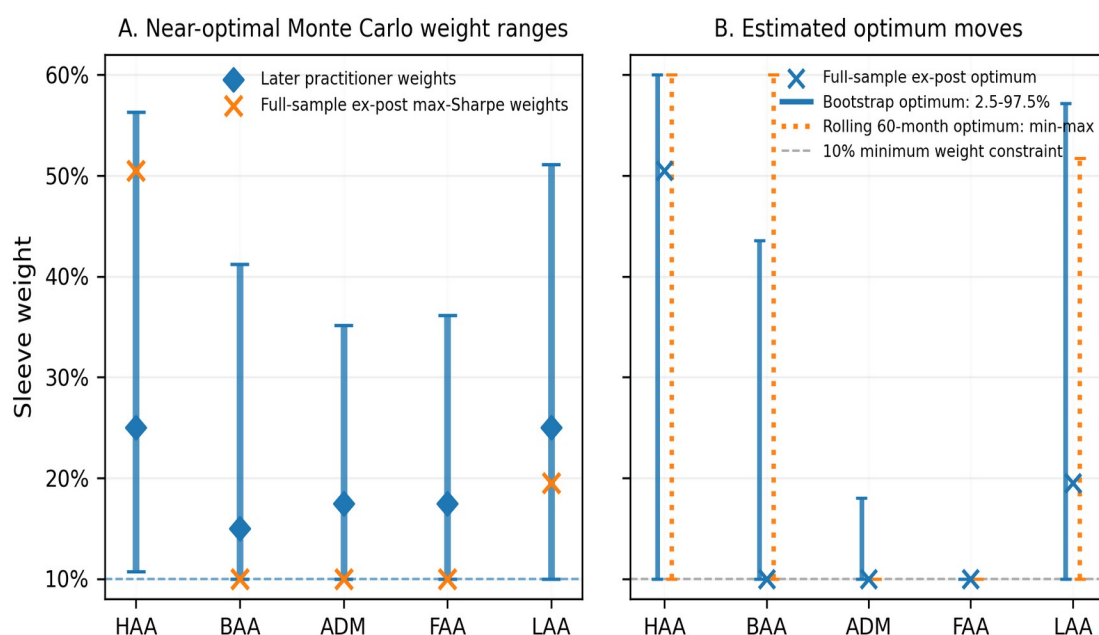


Figure 5. The near-optimal weight region is broad, while the estimated optimum moves. Panel A shows the observed minimum and maximum sleeve weights among the 40,329 of 100,000 feasible Monte Carlo portfolios that retain at least 95% of the full-sample maximum Sharpe. These are sample ranges, not exact boundaries of the feasible set. Panel B shows block-bootstrap 2.5-97.5% ranges (solid blue) and rolling 60-month min-max ranges (dotted orange) for the ex-post optimum. When a rolling range collapses to the 10% floor, it appears as a short orange cap rather than a vertical interval. The dashed horizontal line is the 10% minimum-weight constraint.

The lesson is not to reject optimization. It is to reject confidence in one precise historical optimum when materially different allocations perform similarly and the estimated optimum moves with the sample.

Takeaway. *The relevant question is the stability of the near-optimal feasible region, not the location of one historical optimum.*

That is a design principle, not a statistical guarantee. The evidence here says only that observed ADAA performance is relatively insensitive across a broad range of sleeve weights while the estimated optimum moves substantially.

A post-hoc robustness check reaches the same practical conclusion. A rolling 60-month maximum-Sharpe rule uses the prior 60 completed holding-period returns, keeps each sleeve between 10% and 60%, and applies the estimated weights in the next month. It earns slightly more gross CAGR but trades more. After

applying a 25-basis-point one-way trading cost, the fixed practitioner allocation has the higher CAGR and Sharpe. Because this check was designed after the main performance results were examined, it is exploratory rather than confirmatory and is not used to replace the fixed allocation. It does not prove that fixed weights always beat rolling optimization; it shows that chasing a moving optimum is not free.

The framework therefore separates two sources of uncertainty:

- **Decision Diversification** addresses uncertainty about which rule will work best.
- **Weight Robustness** addresses uncertainty about how precisely those rules should be combined.

8. What Does Decision Diversification Look Like in the Data?

The primary investable sample runs from June 2008 through July 2026 and uses the ETFs in the ADAA portfolio studied here together with the histories required by its lookback rules. The start date is determined by data availability rather than portfolio performance. Because several public source rules and investable versions postdate June 2008, the evidence is a historical common-sample simulation, not a live or post-publication track record.

The ADAA portfolio using the later practitioner weights produces the following results over the 218-month sample:

Table 2A. Core observed performance and implementation costs

Portfolio	Basis	CAGR	Volatility	BIL-excess Sharpe	Maximum drawdown
ADAA — practitioner weights	Gross	10.80%	8.97%	1.05	-10.34%
ADAA — practitioner weights	25 bps one-way	9.36%	8.97%	0.90	-10.75%
ADAA — equal 20%	Gross	10.66%	8.97%	1.03	-10.00%
ADAA — equal 20%	25 bps one-way	9.07%	8.97%	0.87	-10.43%
HAA — hindsight strongest sleeve	Gross	12.97%	10.70%	1.07	-11.68%
HAA — hindsight strongest sleeve	25 bps one-way	11.20%	10.76%	0.92	-12.30%
60/40 SPY/IEF	Gross	8.44%	9.72%	0.75	-27.55%
60/40 SPY/IEF	25 bps one-way	8.38%	9.71%	0.75	-27.61%
SPY	Gross	11.68%	15.73%	0.70	-46.32%

Cost note: transaction costs are charged on final-account one-way turnover after offsetting sleeve-level trades. SPY is unchanged across the trading-cost assumptions because it is buy-and-hold in this comparison.

The table does not tell a winner-takes-all story. HAA has the highest gross CAGR, and SPY also earns more than ADAA while taking materially more volatility and drawdown. ADAA is not a return-maximizing strategy. Its case rests on the observed risk-return trade-off and on whether that trade-off survives implementation and robustness checks, not on the highest terminal wealth.

Trading costs materially reduce gross performance. We use one-way cost assumptions of 0, 5, 10, 25, and 50 basis points per unit of turnover, set before this comparison. At 25 basis points, ADAA using the later

practitioner weights falls to about 9.36% CAGR and 0.90 Sharpe. The SPY/IEF 60/40 benchmark is about 8.38% CAGR and 0.75 Sharpe under the same assumption. Implementation is part of the result, not a footnote to it.

Table 2B. Uncertainty checks accounting for serial dependence

Comparison	Basis	Annualized mean-return difference	HAC(12) p-value	Bootstrap freq. (mean-return advantage)	Bootstrap freq. (Sharpe advantage)	Bootstrap freq. (MDD advantage)
ADAA vs 60/40	Gross	+2.10 pp	0.205	91.0%	95.2%	97.3%
ADAA vs 60/40	25 bps	+0.83 pp	0.623	67.5%	78.2%	96.0%
ADAA vs HAA hindsight winner	Gross	-2.12 pp	0.057	2.3%	38.5%	86.5%
ADAA vs HAA hindsight winner	25 bps	-1.85 pp	0.111	5.0%	41.4%	84.3%

Note: MDD = maximum drawdown. Bootstrap results use 12-month moving blocks; the reported advantage percentages are resampling frequencies, not p-values or posterior probabilities. An MDD advantage means a shallower maximum drawdown for ADAA.

The block-bootstrap and HAC results broadly point in the same direction. A 12-month moving-block bootstrap shows a high resampling frequency of ADAA having a shallower maximum drawdown than the SPY/IEF 60/40 benchmark, and the gross Sharpe difference is positive in most bootstrap draws. But the HAC test for the average gross return difference is not conventionally significant. The evidence is therefore stronger for risk-adjusted behavior and drawdown than for a statistically significant mean-return advantage.

ADAA does not beat the best constituent in hindsight. HAA is the strongest full-sample sleeve, at roughly 13.0% CAGR and 1.07 Sharpe gross. That is the model-selection problem in plain view. The best rule is easy to name after the fact; the investable problem is that hindsight is not available when the capital has to be allocated.

Rolling evidence makes the distinction clearer. On 60-month rolling Sharpe, ADAA using the later practitioner weights is above the median constituent in about 98.7% of windows. It appears in the top two among the five sleeves plus the ADAA portfolio in about 69% of windows and is never in the bottom two.

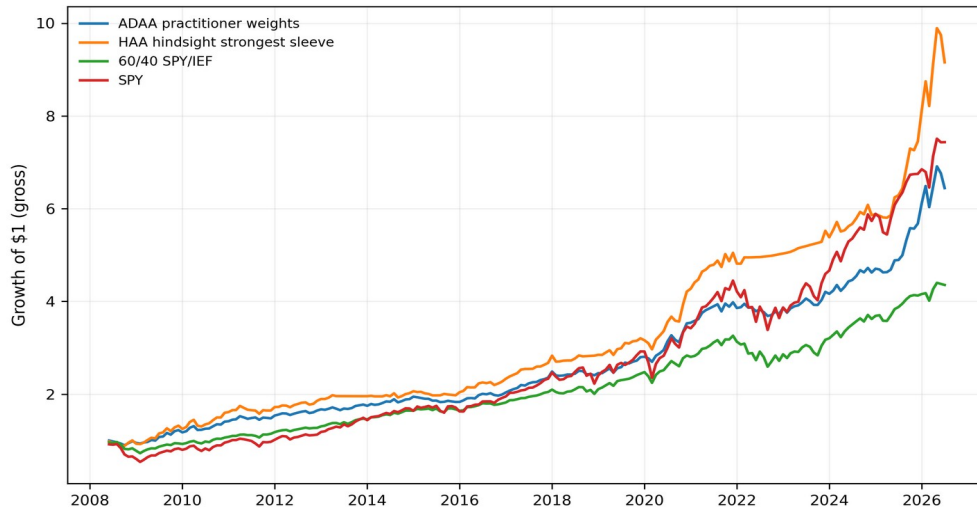


Figure 6. Cumulative wealth makes the trade-off visible. Growth of \$1 is shown for gross ADAA using the later practitioner weights, the hindsight-strongest HAA sleeve, the SPY/IEF 60/40 benchmark, and SPY over the common June 2008-July 2026 sample. The chart is descriptive: it shows the path behind the summary statistics, not a claim that ADAA maximizes terminal wealth.

Drawdown, recovery, and survivable compounding

Drawdown matters because compounding remembers losses. A 20% loss requires a 25% gain to recover; a 50% loss requires a 100% gain. Deep losses therefore create a recovery burden that volatility alone does not show. Research has long treated peak-to-trough loss as a distinct portfolio constraint (Grossman and Zhou 1993; Goldberg and Mahmoud 2017). More recent work treats drawdown duration as a separate dimension of risk (Vedernikov, Liesiö, and Seppälä 2024), while investor-flow evidence suggests that investors respond materially to maximum drawdowns (Riley and Yan 2022).

The goal is not to minimize drawdown at any cost. Too much protection can starve participation in growth, and a shallow-drawdown strategy can still compound poorly. The practical objective is survivable drawdown: losses shallow and recoverable enough that the investor can stay invested and the capital base can keep working.

For consistency, drawdown is measured from the running wealth peak with the initial invested wealth of 1.0 included as a valid starting peak. That convention matters in a sample that begins during the 2008 crisis because an immediate loss should count as a drawdown from the capital initially invested.

Under the standard definition, gross ADAA using the later practitioner weights has a maximum drawdown depth of 10.34%. Recovering from that trough requires an 11.53% gain. The SPY/IEF 60/40 benchmark has a 27.55% maximum drawdown, requiring 38.03% to recover, while SPY falls 46.32% and requires 86.30%. The longest observed underwater spell is 18 months for ADAA, 27 months for 60/40, and 32 months for SPY. These are descriptive sample outcomes, not forecasts of future protection.

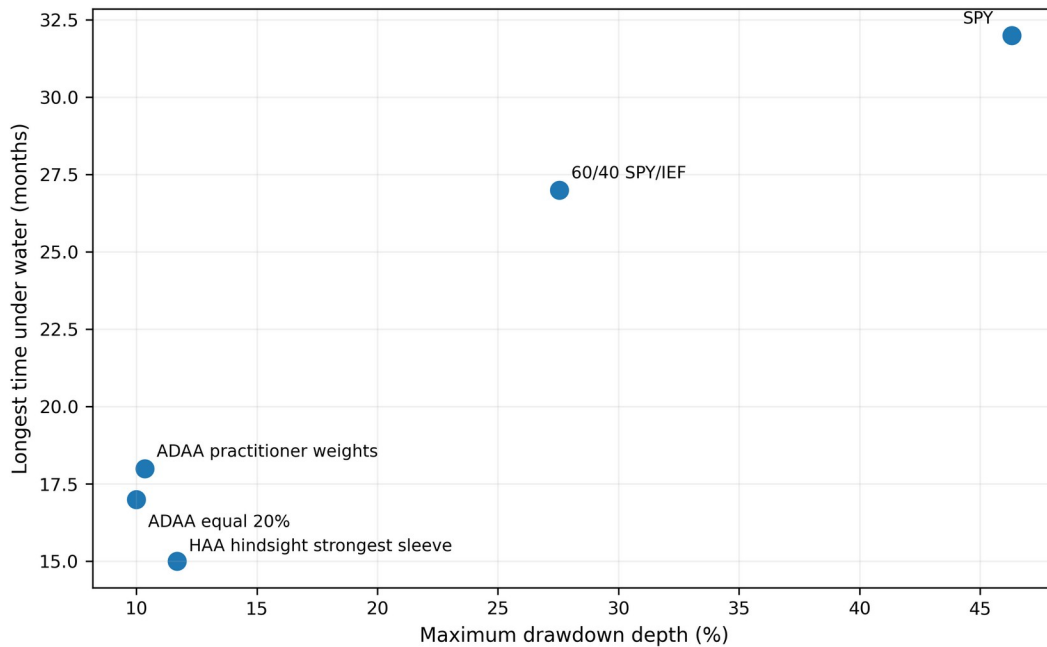


Figure 7. Drawdown has both depth and duration. The horizontal axis shows maximum drawdown depth, and the vertical axis shows the longest consecutive spell below a prior wealth peak. The figure uses the standard definition in which initial invested wealth, $W_0 = 1$, is a valid running peak. A portfolio that falls less deeply can still remain underwater for a meaningful period; neither statistic should be interpreted in isolation.

Table 3. Drawdown anatomy and recovery burden — gross returns

Portfolio	Maximum drawdown	Gain required to recover	Longest underwater run	MDD trough	MDD recovery
ADAA — practitioner weights	-10.34%	+11.53%	18 months	2008-10	2009-05
ADAA — equal 20%	-10.00%	+11.11%	17 months	2008-10	2009-05
HAA — hindsight strongest sleeve	-11.68%	+13.23%	15 months	2008-10	2008-12
60/40 SPY/IEF	-27.55%	+38.03%	27 months	2009-02	2010-09
SPY	-46.32%	+86.30%	32 months	2009-02	2011-02

The comparison prevents an overly simple story. HAA has the higher CAGR and Sharpe and a shorter longest underwater spell than ADAA, even though its maximum drawdown is slightly deeper. Drawdown control is therefore one part of a survivable architecture, not a standalone objective function. Protection has value only if it leaves enough participation for compounding to continue, so drawdown is treated alongside return, volatility, and turnover rather than minimized in isolation.

The role of the slow sleeve

LAA makes another point: mechanism should be separated from headline performance.

Removing LAA and rescaling the other four sleeves slightly increases gross CAGR in an equal-weight comparison. It also raises volatility and turnover. At a 25-basis-point trading-cost assumption, the version with LAA has the higher CAGR and Sharpe in this sample.

But LAA does not universally improve drawdown. Replacing LAA with RAA (Keller 2020), another slow-moving rule, produces an even shallower observed drawdown but slightly lower return. The lesson is not that LAA is the best defensive strategy. It is that the choice of slow-moving sleeve changes the portfolio's overall risk-return profile, and persistence itself is a distinct design dimension.

9. Where It Fails

The evaluation is incomplete without explicit failure modes.

9.1 Rapid reversal and re-entry lag

The clearest recurring failure pattern is a sharp rebound after a weak prior market.

Using a fixed screening rule, we identify the ten worst one-month ADAA returns relative to the SPY/IEF 60/40 benchmark. Seven occur when SPY had fallen over the prior three months and then rose more than 3% in the current month. Examples include March and April 2009, October 2015, January 2019, July and October 2022, and November 2023.

The intuition is straightforward:

Takeaway. *A momentum rule can avoid part of a fall and still miss part of the first rebound.*

That lag is the price of waiting for evidence before taking risk again.

9.2 Not every stress episode favors ADAA

During the portion of the Global Financial Crisis covered by the sample, ADAA falls about **4.5%**, compared with roughly **23.0%** for the SPY/IEF 60/40 benchmark and **41.9%** for SPY. ADAA also loses less than those benchmarks in the COVID crash and in the 2022 stock-bond stress window.

But 2011 is an important counterexample. ADAA loses about 2.0% over the May-October window defined in advance, while SPY/IEF 60/40 is slightly positive. Adaptation is not a guarantee of winning every risk-off period.

Figure 8 summarizes the asymmetry. The bars show cumulative ADAA return minus 60/40 return for the stress windows specified in advance. The final bar is deliberately different: it averages the seven worst relative months that also meet the post-hoc rapid-reversal condition. It is an exploratory failure check, not a stress window specified in advance.

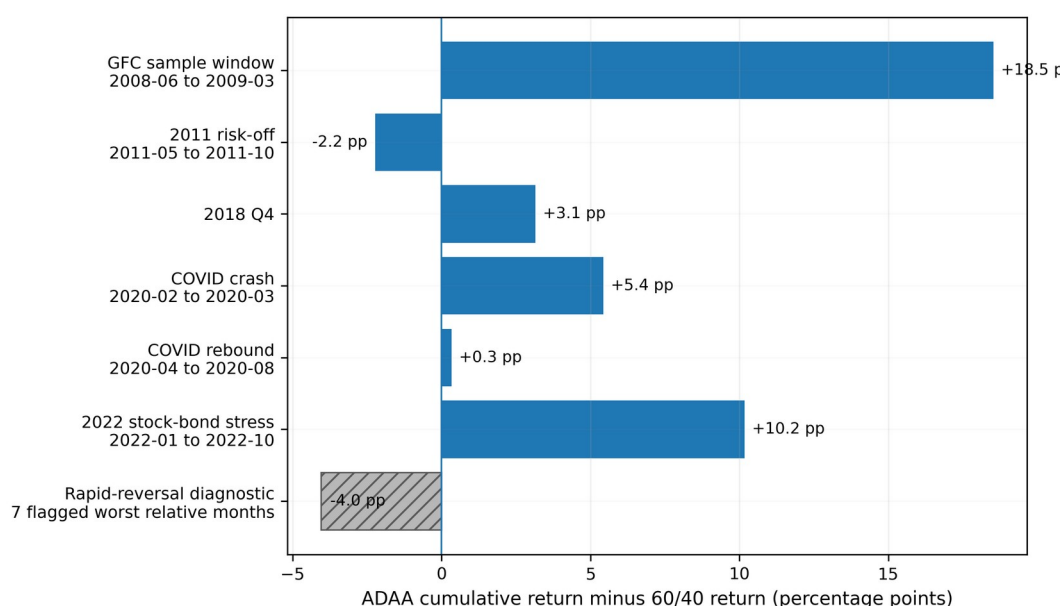


Figure 8. Stress protection is episodic; rapid reversals remain a failure mode. ADAA is substantially ahead of 60/40 in the GFC portion covered by the sample, COVID crash, 2018 Q4, and 2022 stock-bond stress, but not in 2011. The hatched final bar is the exploratory rapid-reversal diagnostic rather than a prespecified stress window; it averages seven of the ten worst one-month relative outcomes and is negative by about four percentage points.

9.3 Slow does not mean safe

LAA loses about **19.6%** in the 2022 stock-bond stress window, even though HAA is close to flat and BAA is positive over the same episode.

Slow is a clock, not a guarantee. Sometimes persistence helps; sometimes it means holding the wrong thing for longer.

9.4 Decision diversity does not eliminate common market risk

No stress episode defined in advance has all five sleeves changing target weights simultaneously. That is evidence of decision disagreement, but not immunity from loss. Different decisions can still produce losses together when the underlying asset opportunity set is broadly unfavorable.

Decision Diversification reduces concentration in the decision process; it does not remove common market risk.

10. Rules for a Portfolio That Must Evolve

The current sleeves and their number may change. The questions below should still apply. That is the point of treating ADAA as an architecture rather than a fixed recipe.

1. Require a rationale before looking at backtest performance. Economic, behavioral, or implementation logic should come first; backtest results can then serve as supporting evidence.
2. Assess decision diversity directly. Strategy labels are not evidence of diversification; evaluate what is held, when allocations change, and how strongly risk is expressed.
3. Use return correlation, but do not stop there. Add holdings overlap, target-weight distance, transition timing, defensive exposure, concentration, and holding persistence.

4. Diversify response horizons. If rules de-risk and re-enter on similar schedules, several strategies can still constitute one timing exposure.
5. Treat investable substitutions as model changes until their effects are demonstrated. A broader universe or replacement ETF can improve implementation while altering the decision role.
6. Do not assign a slow-moving sleeve the role of universal hedge. Persistence can diversify timing without guaranteeing drawdown protection.
7. Evaluate the stability of the near-optimal weight region. Prefer allocations whose performance remains similar across materially different feasible weights rather than relying on one estimated optimum.
8. Measure implementation costs at the final-account level. Net offsetting underlying-asset trades across sleeves before computing realized portfolio turnover.
9. Identify recurring failure patterns. Rapid rebounds after defensive shifts, synchronized de-risking, and persistent exposure to an unfavorable regime are more actionable than a generic list of losing months.
10. Set sleeve-review and replacement criteria in advance. Retain, revise, add, remove, or replace a sleeve using documented tests of role, redundancy, robustness, and implementation feasibility. Define the criteria tightly enough to limit after-the-fact rewriting.

11. Korea-Based Investor Extension: Currency Exposure

Because ADAA was developed from the perspective of a KRW-based investor, the USD/KRW exposure problem is practically important even though it sits outside the core Decision Diversification claim.

The historical currency rule uses a long-horizon USD/KRW z-score to vary the unhedged USD share among 90%, 50%, and 10%. The thresholds and 1,306-trading-day lookback are left unchanged from the historical rule rather than retuned after seeing the present sample.

Table 4. USD/KRW currency-exposure comparison - gross underlying ADAA path

Currency exposure	CAGR	Volatility	Maximum drawdown
Hedged proxy	10.80%	8.97%	-10.34%
Fixed 50% unhedged	11.99%	8.41%	-9.39%
Fully unhedged	12.79%	11.57%	-14.32%
Legacy dynamic 90/50/10	13.25%	8.17%	-6.93%

Note: These are historical currency-exposure outcomes on the same gross underlying ADAA path, not a forecast or an executable hedge comparison. The full appendix also reports Sharpe, recovery burden, underwater duration, and the 25-basis-point underlying-cost rows.

The comparison is straightforward. The legacy 90/50/10 path has the highest CAGR, lowest volatility, and shallowest maximum drawdown among the four variants, while full USD exposure raises return and risk together. Appendix FX Figure 1 shows the full cumulative wealth paths. The result is historical application evidence, not a claim that the legacy currency rule is ex-ante optimal.

The sensitivity checks matter because the historical thresholds were chosen with past outcomes in view. All sixteen threshold combinations defined in advance have a higher observed CAGR than fixed 50% unhedged in this sample, and fifteen have a shallower maximum drawdown.

The historical -0.5/+2.0 pair is not the ex-post CAGR maximum, and the 756-, 1,306-, and 1,827-trading-day windows give similar results on their common evaluation period. That makes the result look less dependent on one narrowly tuned setting, but it does not turn a historically chosen rule into a claim about future performance. Appendix B reports the full check.

What the currency analysis does not include

The 'hedged' leg is an exposure proxy, not a traded hedge. It does not incorporate USD/KRW forward points, NDF bid-ask spreads, collateral or funding costs, hedge-rebalancing costs, or tax and account-specific effects. The appendix therefore treats the exercise as a currency-exposure extension rather than an implementable hedged-return backtest. Using the same initial-wealth convention ($W_0 = 1$) as in the core analysis, the hedged-proxy maximum drawdown is 10.34%; the monthly return path is unchanged.

Accordingly, the extension is best interpreted as an applied sensitivity analysis of currency exposure, not as an implementable hedging strategy or as additional evidence for the core Decision Diversification claim.

12. Limitations and Boundaries

Sample and timing

The paper uses ETF-only investable data rather than extending history with weaker proxy series. That keeps the simulation closer to an investable portfolio but leaves about eighteen years of common evidence. The June 2008 start is a common-data simulation start, not the beginning of a live ADAA record, and several source rules or variants were published later. Accordingly, all 2008-2026 headline results are historical simulations of the portfolio rules studied here, not post-publication out-of-sample performance.

Strategy-family challenge

The historical families were not selected from an exhaustive census of strategies available in 2023. Appendix A compares 16 public rules documented around the earlier project. Within that set, the historical five-family combination ranks 735th of 4,368, around the 83rd percentile. A different combination has the highest Decision-Space score, and BAA Aggressive and BAA Balanced are the closest pair. Four rules were added only after the Decision-Space definition had been fixed, so they provide a limited check on an already specified score. This is not historical preregistration, out-of-sample validation in market time, or proof that the Decision-Space measure is uniquely correct. An independent cross-check reproduced the added rules' target weights to numerical precision without using performance data.

From public rules to investable sleeves

The investable sleeves are not untouched copies of their public parents. Comparisons between public parent rules and investable versions make those changes visible, but they cannot eliminate specification or data-mining risk. They also reveal a practical tension: improving diversification within a sleeve can make the sleeves more similar to one another. The present five-sleeve set should therefore be read as one empirical example of a portfolio design that can evolve, not as permanent components or a target count.

Data, costs, and FX

LAA's unemployment input is aligned to the historically available information set, including the missing October 2025 observation, but the analysis uses the current data vintage rather than a full ALFRED-style first-release history. Trading costs are evaluated under several one-way cost assumptions. Taxes, market impact, fund closure risk, and investor-specific constraints remain outside the model. The USD/KRW appendix studies currency exposure rather than an executable hedge because it does not model forward/NDF carry, spreads, funding, collateral, hedge-rebalancing costs, or taxes.

Claim boundary

Decision Diversification is a practitioner label, not a claim that the underlying idea is entirely new. Forecast diversity, model averaging, signal diversification, strategy ensembles, model-speed diversification, and

position-overlap analysis are relevant precedents. The contribution here is a DAA-specific operating discipline. It asks managers to compare What, When, and How Much, and to track transition timing and persistence. Managers should also check whether each sleeve still serves a distinct role, test whether results depend on precise sleeve weights, and use explicit criteria when sleeves are reviewed or replaced. Public replication materials support the empirical results.

13. Conclusion: What Should Stay When the Strategies Change?

Momentum is useful not because it reveals the next regime, but because it offers a disciplined way to adapt when market evidence changes. Yet one successful idea can be expressed through many reasonable lookbacks, universes, defensive conditions, and portfolio rules. A portfolio can therefore contain several strategies and still depend on much the same decision logic.

This leads to the paper's central proposal: diversification should extend to the decision process, not stop at the asset list. What a rule owns, when it changes, and how much risk it takes can reveal differences that return correlation misses. Comparing public source rules with their investable versions adds an important complication: improving diversification inside a sleeve can make sleeves behave more alike. Adaptation inside each sleeve and diversification across sleeves therefore have to be considered together.

The empirical record does not support a winner-takes-all conclusion. The later practitioner weights are not treated as weights chosen in advance for this study or as optimal. In the present sample, however, they sit within a broad near-optimal region while estimated optimal weights move substantially across rolling windows and bootstrap samples. ADAA does not beat the best individual sleeve in hindsight, and it fails in identifiable environments such as rapid reversals after defensive shifts. The evidence does not establish guaranteed outperformance, a permanently superior sleeve set, or a timeless rulebook.

The more durable conclusion concerns portfolio governance. A multi-rule architecture should diversify decision processes, test whether good results persist across a broad range of sleeve weights, and change components through documented criteria rather than chasing past performance. Neither the identities nor the number of sleeves is fixed; the component count is an implementation choice, not a design target.

A robust allocation architecture should require fewer things to be exactly right at once. The next regime, the future winning rule, the precise weight vector, the number and identity of useful components, and the timing of the next transition are all uncertain.

Drawdown management matters because deep or prolonged losses can interrupt compounding, but minimizing drawdown is not the sole objective. The broader objective is a transparent, implementable process that can absorb uncertainty, continually reassess its components, and evolve without claiming to eliminate uncertainty.

Appendix A. Comparing 16 Public Rules in Decision Space

A fair criticism of ADAA is that several sleeves are adapted from familiar public allocation strategies. The right response is not to hide those source histories. It is to ask whether the portfolio contains genuinely different decisions and to allow the test to disagree with the historical design.

We compare 16 public rules associated with the 2023 design on a common data history: AAA, ADM, BAA Aggressive, BAA Balanced, DAA, DM, EAA, FAA, GPM, GTAA, KDA, LAA, PAA, QSF, RAA, and VAA. For rules whose 2023 implementation followed the DB Financial Investment asset-allocation report, we reconstruct the rule logic from the operational recipes cited by the June 2023 project (Kang 2022).

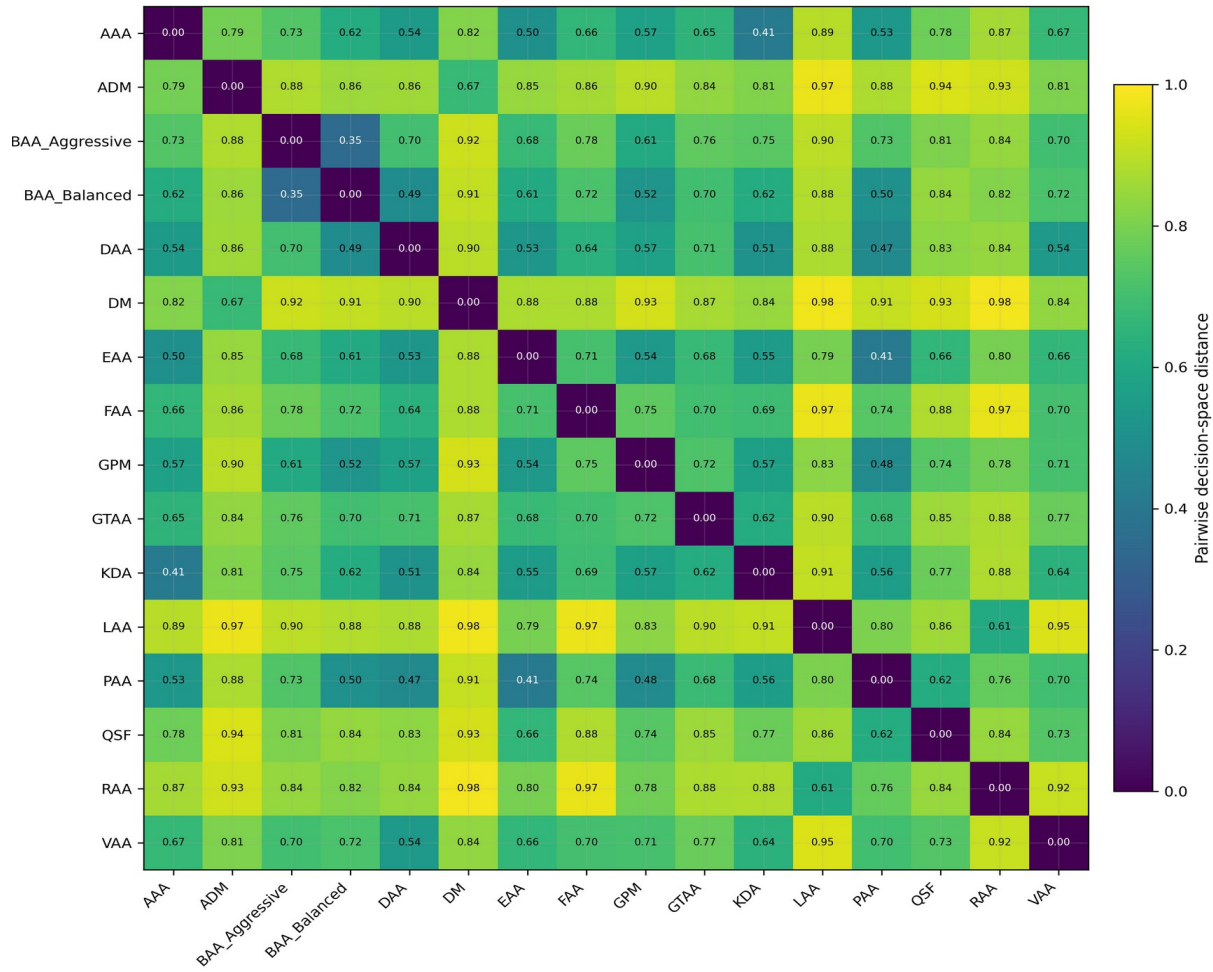
For each pair of rules, Decision-Space compares three parts of the portfolio decision. What: half the average sum of absolute target-weight differences after equivalent ETF exposures are mapped to common economic buckets. Holdings: one minus the average Jaccard overlap of holdings - the share of the combined holdings that the two rules have in common. When: one minus the Jaccard overlap of target-change dates - the share of combined change dates that occur in both rules. Each component runs from 0 (the same) to 1 (maximally different). The pairwise Decision-Space distance is the simple average of What, Holdings, and When. A five-rule set contains ten pairs, so its overall diversity score is the average of those ten pairwise distances. Returns, Sharpe ratios, drawdowns, and other performance statistics are not used. A higher score therefore means more disagreement in portfolio decisions under this specific measure; it does not mean higher expected returns or a statistically better portfolio. All Decision-Space scores in this appendix are computed over the common 216-month target-weight history from July 2008 through June 2026. "Historical 2023 set" refers to the documented five-family composition in the 2023 project, not to a score estimated using only information available at that date.

The Decision-Space definition was set before DM, AAA, PAA, and KDA were added to the 16-rule comparison. Because these four rules were added later, they provide a limited check on a score that had already been specified.

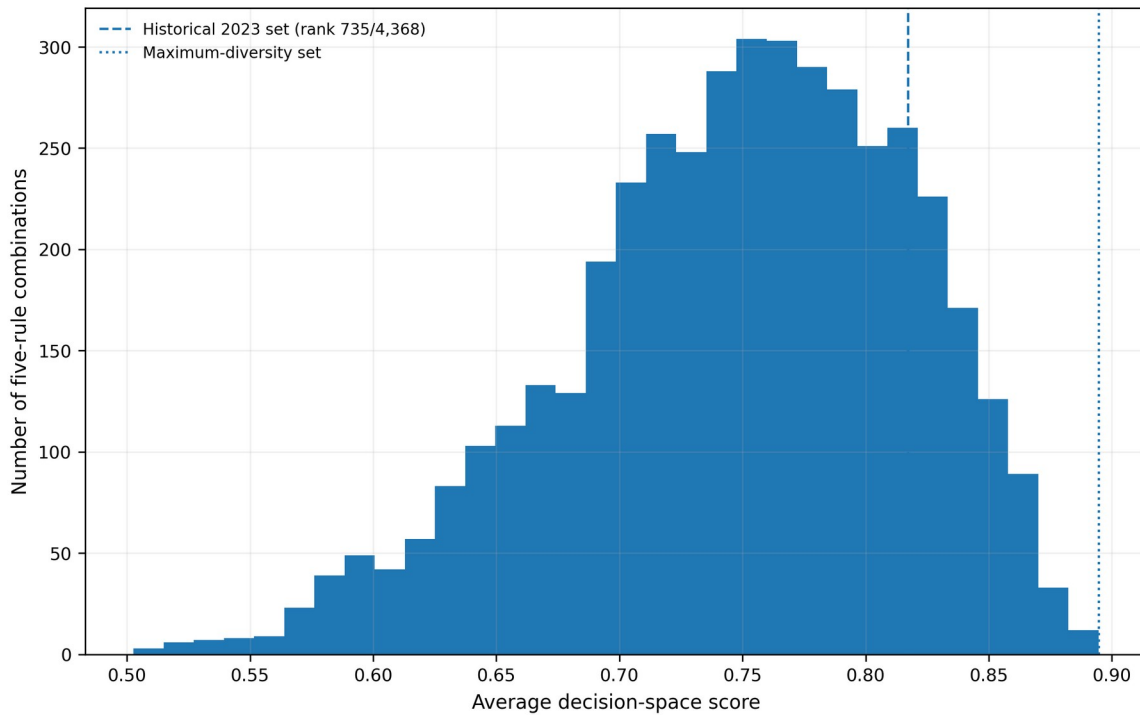
This is not historical preregistration, independent validation of the metric itself, or an out-of-sample test in market time: the market history and the earlier ADAA design were already known.

Among all 4,368 possible five-rule combinations, ADM + DM + FAA + LAA + QSF has the highest Decision-Space diversity score, about 0.895. The historical 2023 family set scores about 0.817 and ranks 735th, around the 83rd percentile. Under this measure, the historical set is more decision-diverse than most feasible sets, but it is not the maximum-diversity set and was not selected by this algorithm.

The most informative challenge is that BAA Aggressive and BAA Balanced are the closest pair among the 16 rules. The return-free comparison therefore identifies redundancy inside the original design rather than simply endorsing it.



Appendix Figure Z1. The 2023 Strategy Zoo in Decision Space. Lower values mean more similar portfolio decisions; higher values mean greater disagreement in target allocations, holdings, and transition dates. No strategy-performance statistic enters the matrix.



Appendix Figure Z2. Historical ADAA was highly decision-diverse, but not maximally diverse. Bars count the 4,368 feasible five-rule combinations by Decision-Space score. Higher scores mean greater decision disagreement, not better returns; the figure is not a performance test.

The result is reasonably stable across alternative weightings of the three Decision-Space components that were set in advance. The historical set ranks between 622 and 1,040, about the 76th to 86th percentile. It stays above the median in every specified variant, but none uniquely selects historical ADAA. This supports the broad finding without showing that one particular distance formula is uniquely correct.

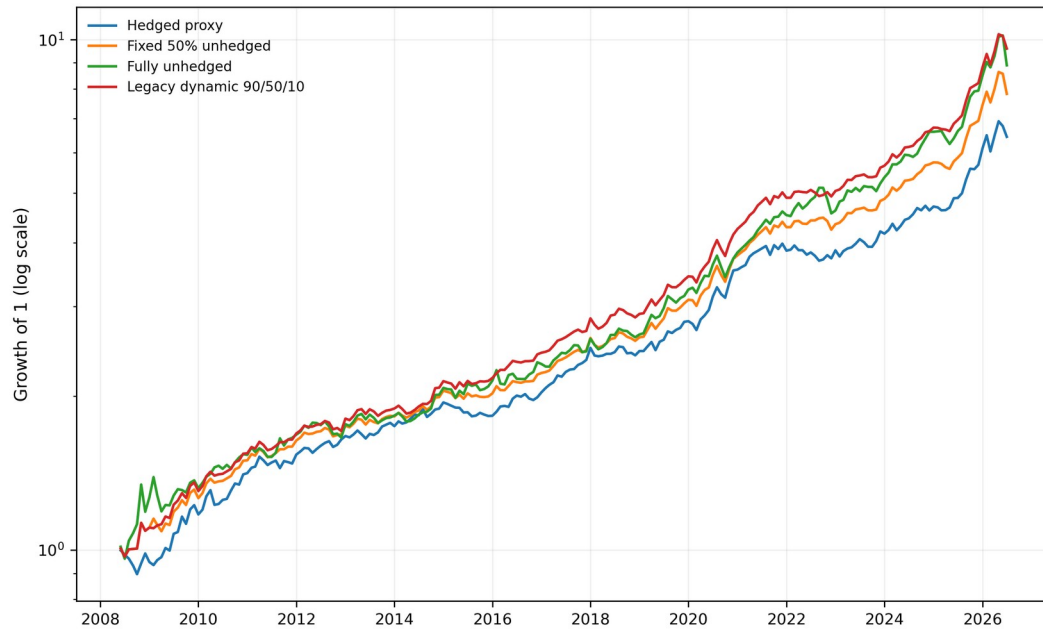
HAA remains outside this historical check because it was not among the rules documented in the June 2023 report. It was already publicly available: Keller and Keuning posted HAA in February 2023.

A separate exploratory comparison adding HAA places the five source-rule families used in the current portfolio around the 96th percentile of a 17-rule set. That result is not used to invent an undocumented historical reason for replacing BAA Balanced with HAA.

An independent cross-language reproduction matched the monthly target weights for DM, AAA, PAA, and KDA to machine precision (maximum absolute difference $< 3 \times 10^{-15}$). The reproduction used no performance data and left the Decision-Space ranking rule unchanged. The public replication materials document this cross-language check.

Appendix B. USD/KRW currency-exposure extension

This appendix isolates the home-currency application from the core global Decision Diversification claim. The historical currency rule uses the 1,306-trading-day USD/KRW z-score used in practice and maps it to three unhedged USD states: 90%, 50%, and 10%. The thresholds are kept at their historical settings rather than reselected from the present results.

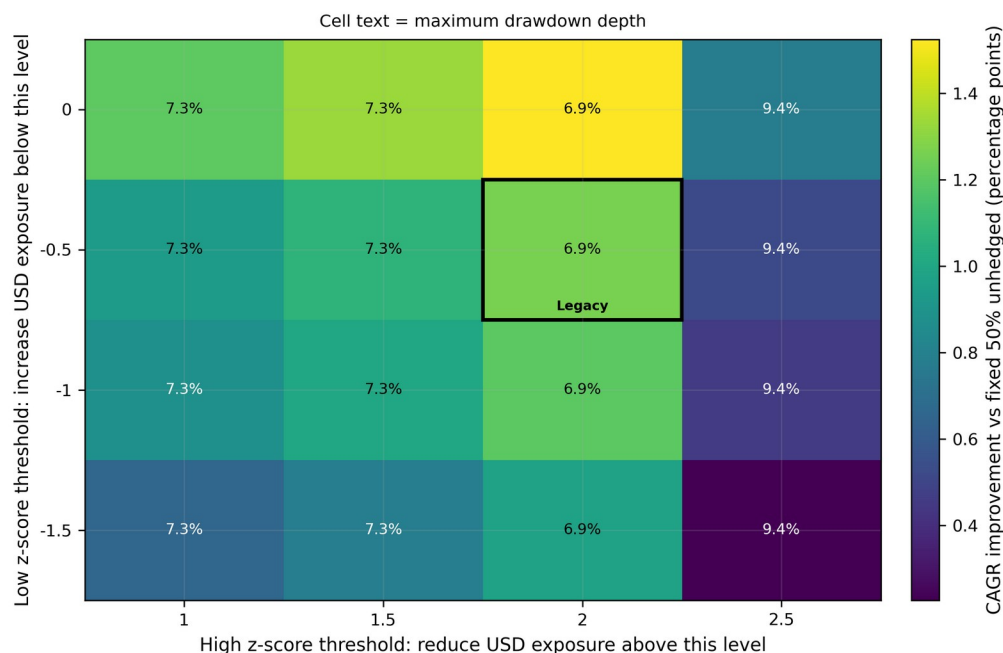


Appendix FX Figure 1. USD/KRW currency-exposure variants. The figure compares wealth growth from an initial value of 1 under the gross underlying ADAA path with four currency-exposure choices. The 'hedged' path is a zero-carry and zero-hedge-cost proxy. It is not an executable forward or NDF return.

Appendix Table FX-1. USD/KRW currency-exposure performance summary

Basis	Currency exposure	CAGR	Volatility	Zero-rate Sharpe	Max drawdown	Gain required to recover	Longest underwater run
Gross underlying	Hedged proxy	10.80%	8.97%	1.19	-10.34%	11.53%	18
Gross underlying	Fixed 50% unhedged	11.99%	8.41%	1.39	-9.39%	10.36%	12
Gross underlying	Fully unhedged	12.79%	11.57%	1.10	-14.32%	16.71%	12
Gross underlying	Legacy dynamic 90/50/10	13.25%	8.17%	1.57	-6.93%	7.45%	7
25 bp underlying cost	Hedged proxy	9.36%	8.97%	1.05	-10.75%	12.05%	23
25 bp underlying cost	Fixed 50% unhedged	10.53%	8.40%	1.24	-9.46%	10.45%	16
25 bp underlying cost	Fully unhedged	11.32%	11.56%	0.99	-14.49%	16.95%	16
25 bp underlying cost	Legacy dynamic 90/50/10	11.78%	8.16%	1.41	-7.03%	7.56%	12

Note: The 25-basis-point rows use the same final-account transaction-cost assumption as the core analysis; no additional currency-hedging implementation cost is modeled. Zero-rate Sharpe is reported only as a descriptive comparison and is not proposed as a Korean-investor benchmark; it should not be compared directly with the core paper's BIL-excess Sharpe.



Appendix FX Figure 2. Threshold sensitivity is broad rather than narrowly tuned. Cell color shows the observed CAGR improvement over fixed 50% unhedged, while cell text shows maximum drawdown depth. The historical -0.5/+2.0 pair is outlined but not presented as optimal. Across the 16 threshold combinations specified in advance, observed gross CAGR ranges from 12.22% to 13.51%, and maximum drawdown depth ranges from 6.93% to 9.39%.

The long-window sensitivity check produces a similarly narrow range of gross CAGR estimates on the common post-warm-up sample. Gross CAGR is approximately 12.45% to 12.59% across 756, 1,306, and 1,827 trading days, and each has a maximum drawdown depth of 6.93%.

In a paired 12-month block bootstrap that preserves within-block serial dependence, the legacy dynamic rule has a positive CAGR difference versus fixed 50% in 99.76% of draws. The dynamic rule has a shallower maximum drawdown than fixed 50% unhedged in about 83.2% of draws. These are resampling checks for this historical application; they do not show that the currency rule will dominate in the future.

Two limitations matter. First, the historical thresholds were chosen with past outcomes in view, so the performance of nearby settings is more informative than the location of the best cell. Second, an implementable Korean hedging study would require forward/NDF carry, spreads, funding/collateral, hedge-rebalancing costs, taxes, and account-specific constraints. Those ingredients remain outside the present paper.

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Data and Code Availability

Public replication materials accompany this working paper. They include research code, metadata for public data sources, derived exhibit data, figures, and concise reproducibility notes. Raw Yahoo/FRED downloads, private university materials, proprietary research, and third-party article files are not redistributed; instructions for retrieving public inputs are included.

Research dashboard: <https://slackquant80.github.io/adaa-slackquant/>

Public replication repository: <https://github.com/slackquant80/adaa-decision-diversification>

Archived replication release (v1.0.2): <https://doi.org/10.5281/zenodo.21853533>

Generative AI disclosure. During preparation of this manuscript, the author used OpenAI’s ChatGPT for limited technical and editorial assistance. The author independently conducted and verified the analysis and takes full responsibility for the content of the manuscript.